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AI-DRIVEN CYBER SECURITY

YOUR AI MIGHT PASS EVERY NORMAL TEST AND STILL FAIL WHEN FACING A REAL ATTACKER.

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WHAT IS AI RED TEAMING?

- A systematic security testing approach for AI systems.
- Simulates real-world attacks to discover vulnerabilities.
- Tests how AI behaves under malicious or unexpected inputs.
- Focuses on identifying weaknesses before attackers exploit them.

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AI RED TEAMING VS TRADITIONAL SECURITY TESTING

- Traditional security testing focuses on applications, networks, APIs, and infrastructure.
- AI Red Teaming focuses on model behavior, prompts, context, outputs, tools, and decision-making.
- AI systems introduce unique attack surfaces that traditional testing may miss.

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UNDERSTANDING THE ADVERSARIAL MINDSET

- Think like an attacker, not a normal user.
- Question assumptions and security boundaries.
- Explore unexpected inputs and edge cases.
- Look for ways to bypass restrictions and manipulate model behavior.
- Test the system beyond its intended use cases.

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WHY NORMAL TESTING IS NOT ENOUGH

- Normal testing checks expected user behavior.
- Attackers deliberately try unexpected and malicious scenarios.
- An AI model can perform perfectly in normal situations but fail under adversarial inputs.
- Security requires testing both intended and unintended behavior.

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UNDERSTANDING AI ATTACK SURFACES

- Prompt Injection and Jailbreaking.
- Sensitive Information Disclosure.
- System Prompt Leakage.
- Tool and Function Calling Abuse.
- RAG and Knowledge Base Manipulation.
- Insecure Output Handling.
- Model and API Endpoint Manipulation.

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SYSTEMATIC PROBING OF ADVANCED MODEL ENDPOINTS

- Identify the AI endpoint and its capabilities.
- Understand its inputs, outputs, tools, and access boundaries.
- Create controlled adversarial test cases.
- Observe model responses and unexpected behavior.
- Document vulnerabilities and assess their impact.
- Repeat testing with different attack variations.

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AI RED TEAMING METHODOLOGY

- RECONNAISSANCE: UNDERSTAND THE TARGET AI SYSTEM.
- THREAT MODELING: IDENTIFY POTENTIAL THREATS AND ATTACK PATHS.
- ATTACK SIMULATION: EXECUTE CONTROLLED ADVERSARIAL TESTS.
- VULNERABILITY ANALYSIS: EVALUATE DISCOVERED WEAKNESSES.
- REPORTING: DOCUMENT FINDINGS, EVIDENCE, AND IMPACT.
- RETESTING: VERIFY WHETHER SECURITY FIXES ACTUALLY WORK.

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RESPONSIBLE AI RED TEAMING

- TEST ONLY SYSTEMS YOU OWN OR HAVE EXPLICIT AUTHORIZATION TO ASSESS.
- USE CONTROLLED ENVIRONMENTS AND NON-SENSITIVE TEST DATA.
- AVOID UNAUTHORIZED ACCESS, DATA EXTRACTION, OR SERVICE DISRUPTION.
- FOLLOW RESPONSIBLE DISCLOSURE PRACTICES.